catalogue

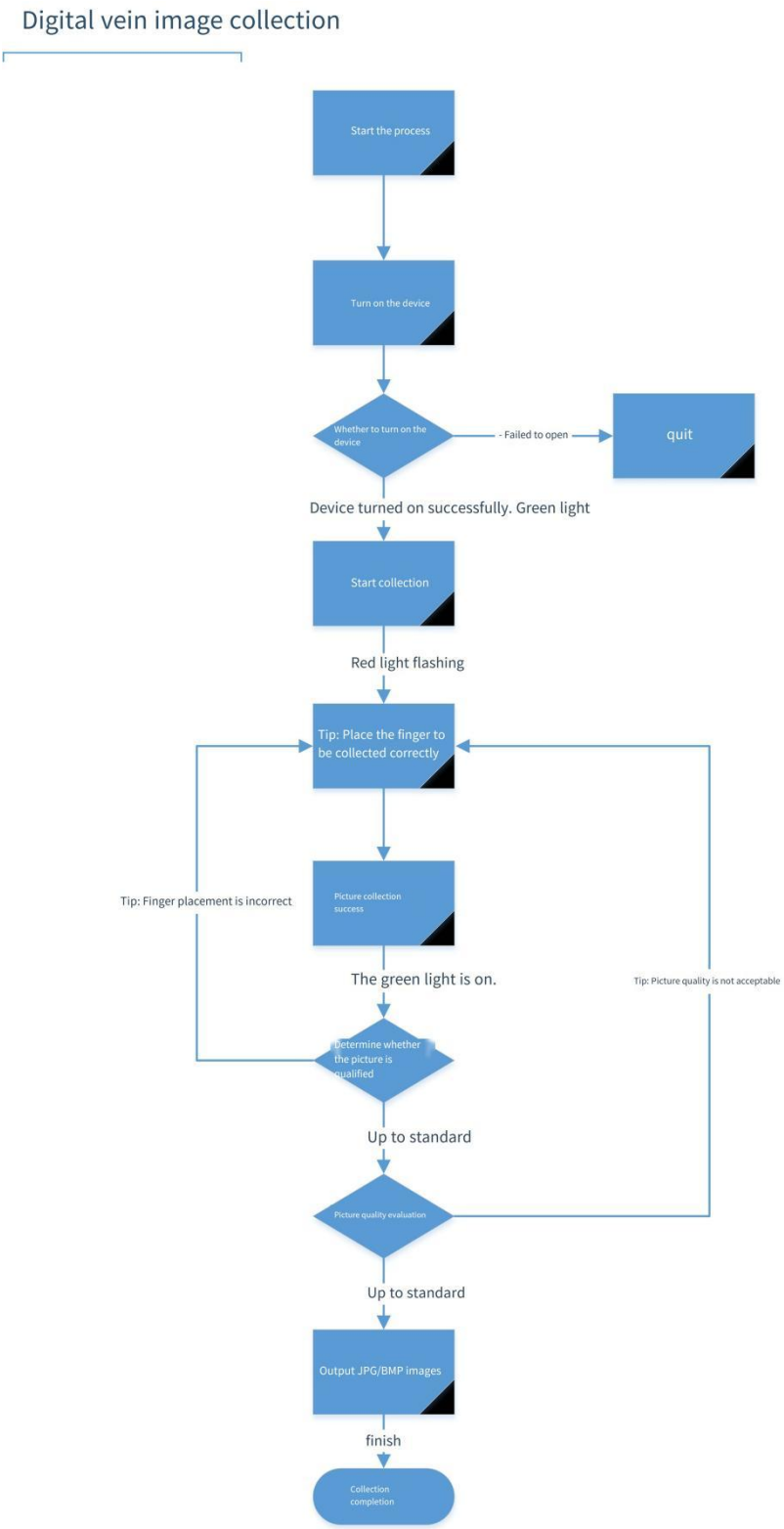
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1. Equipment operation guidelines

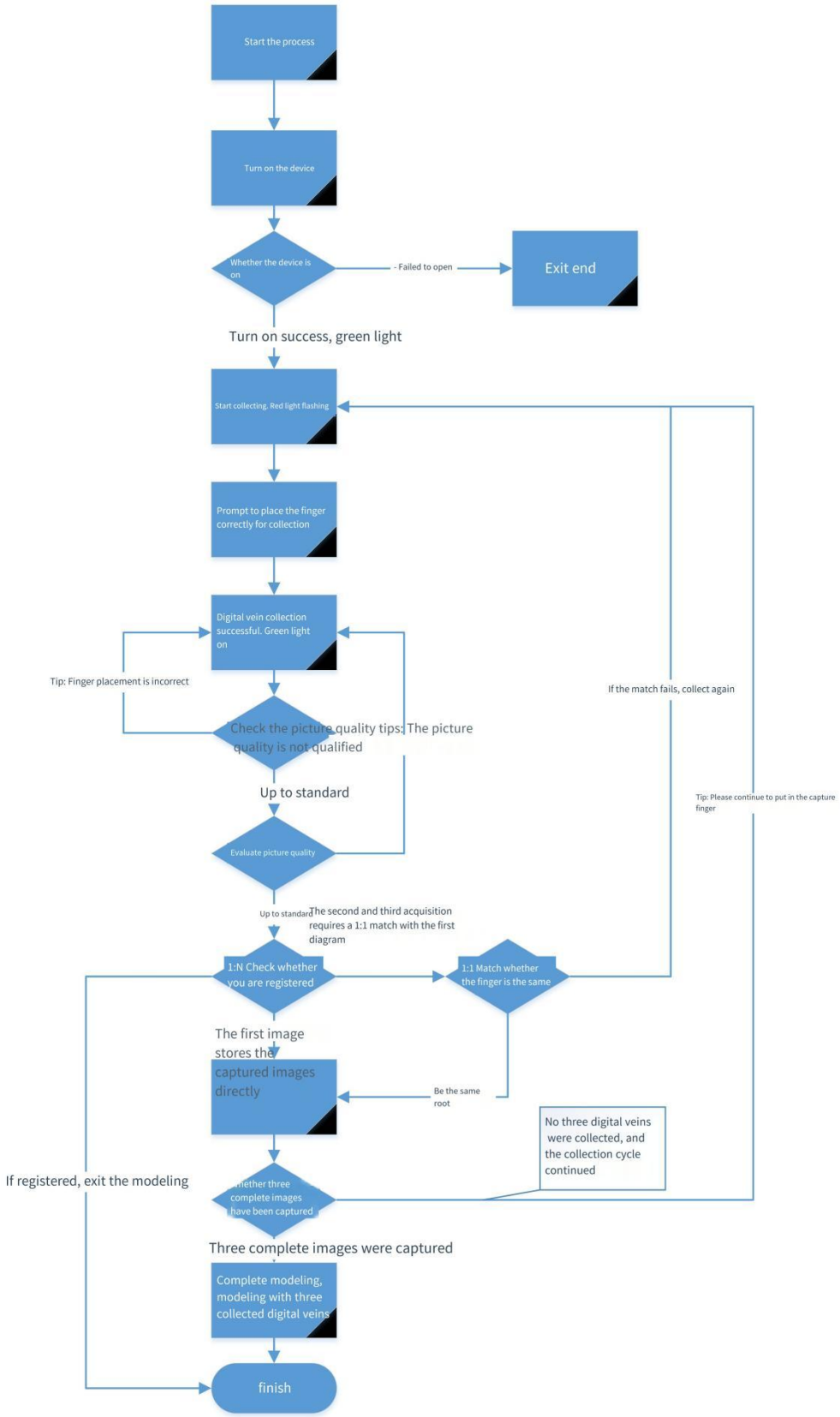
1. Calm down and sit up
2. Dont use your dirty or callused fingers

2. Process introduction

gather

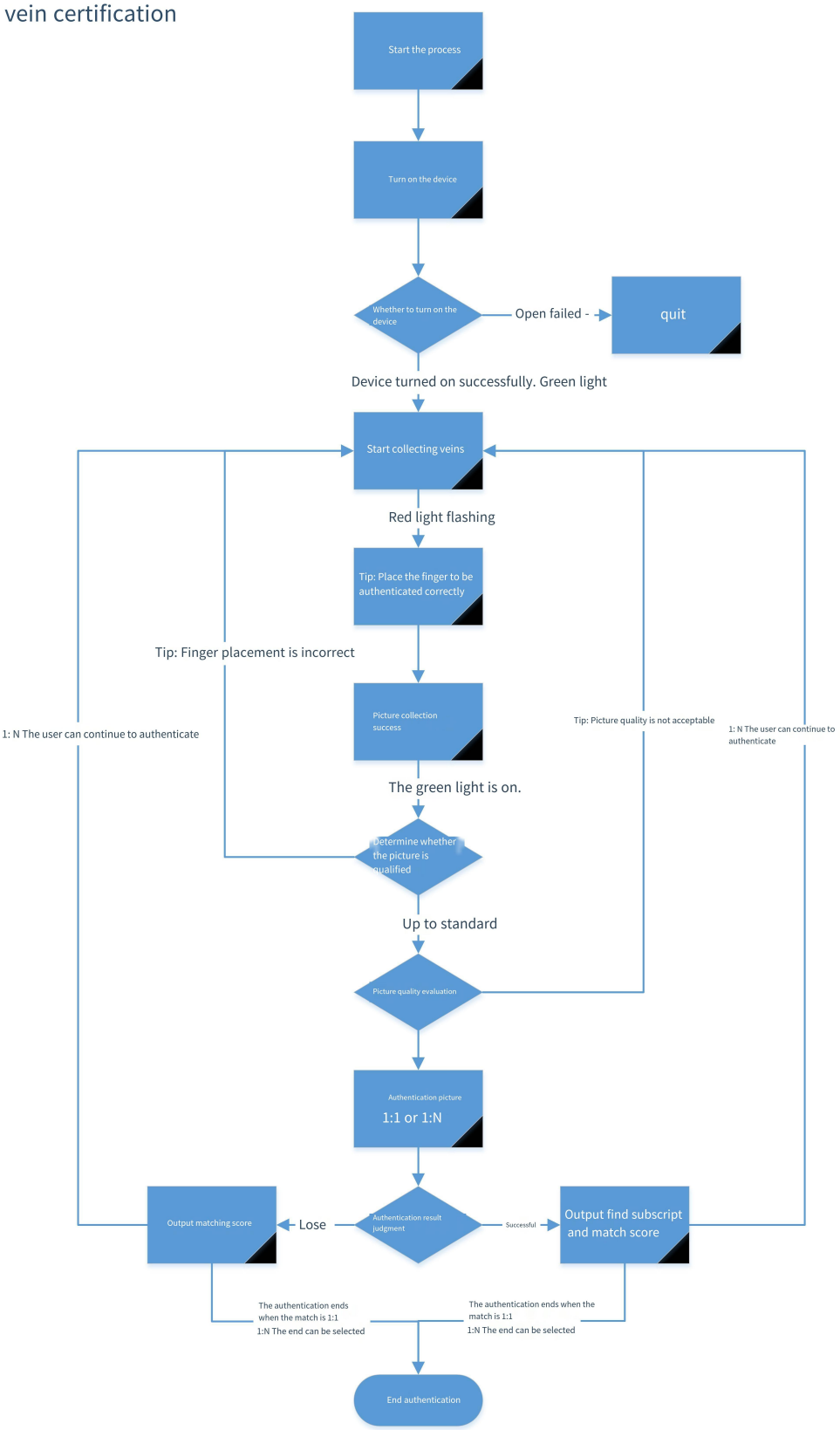


modeling

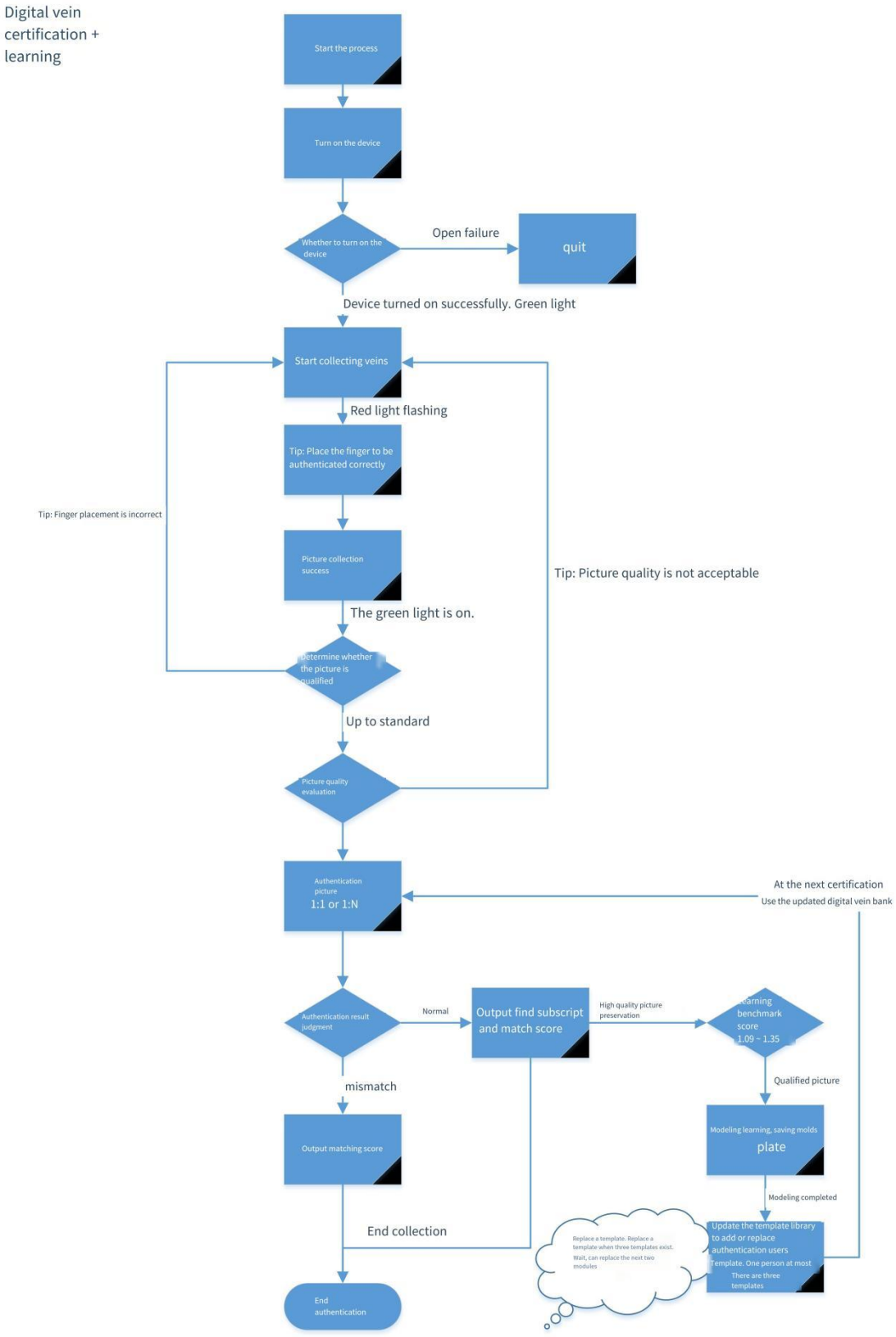

Start the process
Turn on the device
Whether the device is on
- Failed to open
Exit end
Turn on success, green light
Start collecting, Red light flashing
Prompt to place the finger correctly for collection
Digital vein collection successful. Green light on
Tip: Finger placement is incorrect
Check the picture quality tips: The picture quality is not qualified
If the match fails, collect again
Up to standard
Tip: Please continue to put in the capture finger
Evaluate picture quality
Up to standard
The second and third acquisition requires a 1:1 match with the first diagram
1:N Check whether you are registered
1:1 Match whether the finger is the same
The first image stores the captured images directly
Be the same root
No three digital veins were collected, and the collection cycle continued
If registered, exit the modeling
Whether three complete images have been captured
Three complete images were captured
Complete modeling, modeling with three collected digital veins
finish

attestation

Digital vein certification



Certification and learning



3. Function introduction

device initialize

function prototype	**fvdev_open**
Function function	Open the device
devno	Equipment Index (0,1,....)
Hwnd	The return window, in linux is a function of the picture callback, Implementation by the user, typedef void(*pFrameArrived)(unsigned char *pFrame, int width,int height); pFrame: RGB data Width: image width Height: High level of the image

Set the device number

function prototype	fvdev_setno(int no)
Function function	Set the device number
no	number

Get the device number

function prototype	fvdev_ g etno(int *no)
Function function	Get the device number
no	number

The equipment is closed

function prototype	**fvdev_close**
Function function	closing device

Establish a template

function prototype	**fv_extract_model**
Function function	Establish a template
picData1 (Required)	Picture data 1
picLen1 (Required)	Picture data of the 1 in length
picData2 (Optional)	Picture data 2
picLen2 (Optional)	Picture data 2 length
picData3 (Optional)	Picture data 3
picLen3 (Optional)	Picture data of 3 in length
model	Template data
modelLen	The length of the template needs to be passed into a relatively large buffer, at least 4k

Quality assessment

function prototype	BOOL fv_quality(unsigned char *img, int len, int *q)
Function parameters	**Function parameter meaning**
img	Pictures that the img needs to compare
len	Length of the image to compare (length of the img)
q	quality: 1.q=0, indicating mass passing 2.q=2001，Bad Quality with too many pressure 3.q=2003，Light Leak
If the function returns TRUE, the function call is successful; if the function returns FALSE, the function call fails	

1:1 Comparison

function prototype	BOOL fv_match(unsigned char *img, int imgLen, unsigned char*model, float *score)
Function parameters	**Function parameter meaning**
img	Current pictures to compare
model	One of the templates obtained in the registered template
score	Score 052- - -> 10000 0.6- -> 1000000
If the function returns TRUE, the function call is successful; if the function returns FALSE, the function call fails	

1: N comparison

function prototype	BOOL index(unsigned char *models, int count,BYTE*img,int imgLen, int *pos,float *score)
Function parameters	**Function parameter meaning**
model	All of the registered templates
num	The number of registered templates
img	Current pictures to compare
pos	Returns the array subscript for the maximum score
score	Score 052- - -> 10000 0.6- -> 1000000
If the function returns TRUE, the function call is successful; if the function returns FALSE, the function call fails	

Get the number of devices

function prototype	BOOL fvdev_get_count(int *count)
Function parameters	**Function parameter meaning**
count	number

Get finger status

function prototype	BOOL fvdev_get_state(int *state)
Function parameters	**Function parameter meaning**
state	Finger sensing state 0: no finger 1: Fingertip press 2: finger abdomen press 3: finger press

Set the indicator light

function prototype	BOOL fvdev_set_led(int Color, MD_BOOL bFlush)
Function parameters	**Function parameter meaning**
color	1:0: Red: 1: green
bFlush	True: flashing false: always bright

Get a picture

function prototype	**BOOL fvdev_grab(unsigned char *img, int *len)**

Function parameters	**Function parameter meaning**
img	**Allocate a large enough space to pass in, like 30,000 bytes**
len	**The length of the img when incoming and the actual length of the image when outgoing**

Open the device

function prototype	fvdev_open_ex(int devno, HWND hwnd, void **phandle)
Function function	Open the device
devno	Equipment Index (0,1,....)
phandle	Device handle, you can operate a single device separately

gather

function prototype	fvdev_grab_ex(void* handle, unsigned char *img, int *len)
Function function	**gather**
handle	Device handle
img	buffer
len	buffer length

Get finger status

function prototype	fvdev_get_state_ex(void *handle, int *state)
Function function	**Get the hand status**
handle	Device handle
state	Finger state

Set the status of the led lamp

function prototype	fvdev_set_led_ex(void *handle, int Color, BOOL bFlush)
Function function	**Status of the device led lamp**
handle	Device handle
Color	Color (0: Red 1: green)
bFlush	Whether flashing

Get the number

function prototype	fvdev_getno_ex(void *handle, int* no)
Function function	**Get the number**
handle	Device handle
no	number

Set the number

function prototype	fvdev_setno_ex(void *handle, int no)
Function function	**Set the number**
handle	Device handle
no	number

Determine whether the device is turned on

function prototype	isDevOpen_ex(void *handle)
Function function	**Get the device status**
handle	Device handle

Turn off the turned-on device

function prototype	fvdev_close_ex(void *handle)
Function function	**closing device**
handle	Device handle

Get the library version

function prototype	fv_Get_Edition(int *buf)
Function function	**Get the dynamic library version**
buf	Get version parameters

Obtain the venous equipment information

function prototype	fv_Get_Dev_Info(PT_INFO buf)
Function function	**Get device information**
buf	Obtain the device information parameters

Get the venous device number

function prototype	fvdev_getPD(int *a)
Function function	**Get the device number**
a	Return number
function prototype	fvdev_setPD （int a）
function	**Set the device number**
a	The equipment number length shall not exceed 20

Obthe production date of venous equipment

function prototype	fvdev_GetdevNumber(char* devNumber)
Function function	**Get the device date**
devNumber	Return date for example 20170521
function prototype	fvdev_SetdevNumber(char * devNumber)
function	**Set the device date**
a	Format for example 20170521

Get the default alignment threshold value

function prototype	fv_get_default_threshold(floast *score)
Function function	
score	Default threshold
function prototype	
function	

Load the template library

function prototype	BOOL fv_load_model(const char *modelPath, int *num)
Function function	
modelPath	Template library path, the directory structure is: ---(Template id) ----------The specific template file ----------jpg ----------Image file corresponding to the template
num	Number of templates loaded
function prototype	
function	

Search from the template library

function prototype	BOOL fv_user_index(unsigned char *img, int len, int *index, float threshold, float *score,BOOL bUpdate)
Function function	
Img	Images of the finger vein
len	Picture length
threshold	Alignment threshold
score	Maximum score
bUpdate	Whether to update the template
function prototype	
function	

Add templates to the template library

function prototype	BOOL fv_user_add(BYTE *picData1, int picLen1,BYTE*picData2, int picLen2,BYTE*picData3, int picLen3, unsigned int *index)
Function function	
picData1	Refers to venous picture 1
picLen1	Picture length 1
picData2	Refers to venous picture 2
picLen2	Picture length 2
picData3	Refers to venous picture 3
picLen3	Picture length 3
index	If the input is id and 0, the successful registration id is returned If the input id is not 0 and the id is already present in the template library, the return fails, otherwise it fails
function prototype	

function	

Remove the specified id from the template library

function prototype	BOOL MDAPI fv_user_del(unsigned int index)
Function function	
index	user id
function prototype	
function	

Add a new user from the specified folder for synchronization

function prototype	BOOL MDAPI fv_index_addUser(const char *modelPath,int *num)
Function function	**Add users according to the directory**
modelPath	The template storage path where the new user is added
num	The ID number returned after adding, this ID number is consistent with the file name of the storage template
function	

Note: Md_110Seting.ini The file is a configuration file, including the number of storage templates, at least above 2, the default is 5
"Score" is the configuration of the alignment score, not lower than 0.6, the default configuration is 0.6;

4．remarks

Refers to the use instructions and operating procedures for venous equipment
1: Open the device
Obtain the currently available device by using the fvdev _ get _ count function, and the function returns the available standby
Open the device with openDev.
When you open the device, you can use Get_Edition to get the version information of the current library,
Use Get_dev_info to get parameter information for the current device info
2: finger vein modeling
Create a new name, then use getFingerState to get the finger status, under the pressing (0x03) state, use grabImage to get a photo, and then use quality to evaluate the photo quality. After qualified, check if you have registered and use index (1: N).
If it is not registered, please repeat the upper step to get the second photo, and then put the first photo registration template, function getModel. Then compare matchModel (1:1) to compare whether it is the same finger. Resend the above steps, and finally get three photos. Register the template getModel again to complete the finger registration.
3: Finger vein certification
At press (0x03), obtain a photo using grabImage and use quality to assess photo quality. After the photo is qualified, use the index function for 1: N alignment. If the position is found, the output is not found, and the score is lower than the parameter value (0.6).
4: Turn off the equipment
Close the current device by using the fvdev _ close _ ex
5. Please save the pictures of modeling and authentication for subsequent algorithm upgrade and review